

Faculty of Engineering & Technology



Introduction To Data Analytics

CREDIT CARD FRAUD RISK ANALYSIS



Bachelor of Technology in Computer Science

Submitted To:

MR. Harshit Bhargave
Assistant Professor
Faculty of CSED

Submitted By:

Team Name: 03 B.Tech Sec.E
Group Leader: Ayush Sharma
Total member: 12



DECLARATION

I, **Ayush Sharma** hereby declare that the project report titled “Credit Card Fraud Risk Analysis” is an original work carried out by me under the guidance of **MR Harshit Bhargave**. The work presented in this report is based on my learning, analysis, and interpretation using Power BI. The data and information used in the project have been collected from reliable online sources and are used purely for academic purposes. I further declare that this project has not been submitted previously, in part or full, for the award of any degree, diploma, or certificate to any other institute or organization.

Team Members' Contribution:

S. No.	Student Name	Contribution
1	Ayush Sharma (leader)	100%
2	Mukhtar Abbas Rizvi	100%
3	Vipul Kumar Sahu	100%
4	Harikesh Singh	100%
5	Mohammad Saqlain	100%
6	Rameez Raza Khan	100%
7	Ravi Kumar	100%
8	Vansh Rastogi	90%
9	Deepak Kumar	80%
10	Pragyansh Agniotri	80%
11	Pramod	60%
12	Gaurav Sagar	60%

Place:

Date:

Signature of Team Leader

B.Tech (CSE) Sec. E

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CERTIFICATE

This is to certify that the project entitled “**Credit Card Fraud Risk Analysis Dashboard**” is a bona fide work carried out by **Mukhtar Abbas Rizvi** along with the team members of **B.Tech Section E**, during the **academic year 2025–2026**, in partial fulfillment of the requirements for the award of the **Bachelor of Technology in Computer Science (B.Tech CSE)** degree.

The project work has been completed under my guidance and supervision. The team has exhibited outstanding teamwork, technical expertise, and analytical skills throughout the project duration.

I am satisfied with the quality of the work and recommend this project for evaluation.

Mr. Harshit Bhargave

Assistant Professor

Project Guide

External Examiner: _____

Date: _____

Signature: _____

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Team Members:

Ayush Sharma (*Leader*)

Mukhtar Abbas Rizvi

Vipul Kumar Sahu

Harikesh Singh

Mohammad Saqlain

Rameez Raza Khan

Ravi Kumar

Vansh Rastogi

Deepak Kumar

Pragyansh Agniotri

Pramod

Gaurav Sagar

B.Tech (CSE) Section E (Academic Year 2025–2026)

ABSTRACT

Project Title: Credit Card Fraud Risk Analysis

Domain: Data Analytics, Business Intelligence, FinTech, Fraud Detection

Technology Stack: Power BI, Power Query, DAX, Microsoft Excel, CSV Dataset

Abstract

With the rapid growth of **online payments, e-commerce, and digital banking**, the number of **credit card transactions** has increased dramatically. Along with genuine usage, there has also been a significant rise in **fraudulent transactions**, leading to financial losses for banks as well as customers. Traditional manual monitoring systems are often slow, reactive, and unable to handle large-scale data in real time.

This project focuses on designing a **Credit Card Fraud Risk Analysis Dashboard** using **Power BI**. The main objective is to analyze transaction-level data and derive **meaningful visual insights** that help identify suspicious patterns, high-risk customers, risky merchants, and fraud-prone time periods.

The dataset used in this project consists of transaction details such as **transaction amount, time, location, merchant category, transaction mode, and fraud label (fraud / non-fraud)**. The data is cleaned and transformed using **Power Query**, and then modeled in Power BI using **DAX measures** and relationships. Multiple visualizations such as **bar charts, line charts, donut charts, slicers, and KPI cards** are used to represent risk metrics.

The final dashboard allows users to explore fraud trends interactively, compare fraud vs non-fraud transactions, and analyze fraud distribution across **amount ranges, time of day, regions, and merchant types**. This analytical approach supports **data-driven decision-making** for banks and financial institutions and demonstrates how BI tools can improve fraud risk monitoring in a cost-effective way.

Keywords: Credit Card Fraud, Fraud Detection, Risk Analysis, Power BI, Data Visualization, Financial Analytics, Business Intelligence.

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CHAPTER 1: INTRODUCTION

1.1 Background

Digital payments have transformed the way financial transactions are carried out. Credit cards are widely used for **online shopping, bill payments, subscriptions, and POS transactions**. While this convenience has improved user experience, it has simultaneously opened new doors for **fraudsters**. Techniques like **card theft, skimming, phishing, data breaches, and identity theft** are commonly used to commit fraud.

Banks and financial organizations generate **huge volumes of transaction data** every day. If this data is properly analyzed using tools like **Power BI**, it can reveal hidden patterns and help in identifying **high-risk transactions**. Instead of relying only on manual checking or static reports, a **visual dashboard** can provide faster and clearer insights into fraud behavior, which is essential in the era of digital banking.

1.2 Motivation

Credit card fraud results in **financial loss, customer dissatisfaction, and loss of trust** in digital systems. Many frauds go unnoticed until the customer or bank manually detects them, by which time the damage is already done.

The motivation behind this project is:

- To use **data analytics** to better understand fraud patterns instead of guessing.
- To show how **Power BI** can be used as a **fraud monitoring tool**, even without complex coding.
- To build a **clear, interactive dashboard** that can support bank officers in quickly spotting suspicious behavior.
- To gain hands-on experience with **real-world financial data analysis**, which is highly relevant in the **FinTech and IIoT domains**.

1.3 Problem Statement

Although banks collect detailed transaction records, they often lack a **simple and interactive system** to visually analyze fraud trends:

- Data is stored across multiple files and systems.
- Many existing reports are **static** and cannot be filtered dynamically.
- Financial analysts may need to manually check large tables, which is time-consuming.

Problem:

How can we design a **Power BI-based dashboard** that analyzes credit card transaction data and helps in identifying **fraud risk patterns** in an interactive and understandable way?

1.4 Objectives

The main objectives of this project are:

1. To **clean, organize, and analyze** credit card transaction data.
2. To **compare fraud and non-fraud transactions** based on various factors like amount, time, location, and merchant type.
3. To design **KPIs and visualizations** that clearly show fraud trends and high-risk segments.
4. To develop an **interactive Power BI dashboard** with filters and slicers for flexible analysis.
5. To generate **actionable insights** that can support fraud detection strategies and customer risk management.

1.5 Scope of the Project

- The project focuses on **historical transaction data**, not live real-time monitoring.
- It covers **descriptive and diagnostic analytics**, not full machine learning-based prediction.
- The dashboard is designed as a **decision-support visualization tool** for analysts and managers.
- The project is suitable for academic demonstration of **Power BI skills, data cleaning, modeling, and financial analytics**.

1.6 Project Overview

The project begins with **data collection** from a credit card transaction dataset (e.g., publicly available Kaggle dataset or synthetic data). The data is then **cleaned and preprocessed** using Power Query. Next, a **data model** is created in Power BI, with relationships, calculated columns, and DAX measures.

Finally, multiple **reports and dashboard pages** are designed to show fraud vs non-fraud trends, top risky merchants, amount distribution, and time-based fraud behavior. The report ends with **results, conclusion, limitations, and future scope**.

CHAPTER 2: LITERATURE REVIEW

2.1 Existing System Analysis

Existing fraud detection systems in banks generally include:

- **Rule-based systems** (e.g., if amount > X at unusual location, flag).
- **Manual reviews** by fraud analysts going through alerts and reports.
- **Static MIS reports** generated periodically in Excel or other tools.

Limitations:

- Rules may become outdated and **do not adapt** to new fraud patterns.
- Manual checking is **slow, tiring, and error-prone**.
- Data from different channels (ATM, POS, online) may not be **integrated** properly.
- Visual representation is limited, making it hard for managers to quickly see **overall fraud risk**.

2.2 Research Gap

From studying various articles, blogs, and academic papers on fraud detection:

- Most focus on **complex machine learning models**, but ignore **simple visual dashboards** for day-to-day analysis.
- Many solutions are **costly and complex**, not suitable for small banks or educational demonstration.
- There is a lack of **simple, low-cost, Power BI-based analytic dashboards** that allow quick exploration of fraud trends.

This project tries to fill this gap by providing a **clear and easy-to-use Power BI dashboard** for credit card fraud risk analysis.

CHAPTER 3: METHODOLOGY

3.1 Data Collection

- Dataset of credit card transactions with fields like:
 - Transaction ID
 - Customer ID
 - Transaction Date & Time
 - Transaction Amount
 - Merchant Category
 - Transaction Type (Online, POS, ATM, etc.)
 - Country / City
 - Fraud Flag (0 = Non-Fraud, 1 = Fraud)
- Data source:
 - Public dataset (e.g., Kaggle) or synthetic/fake data generated for project purposes.

3.2 Data Cleaning & Preprocessing

Goals of cleaning:

1. Remove **duplicate** transactions.
2. Handle **missing values** (amount, merchant, etc.).
3. Ensure correct **data types** (date, number, category).
4. Standardize **merchant categories and location names**.
5. Add **derived columns** useful for analysis.

Typical steps in Power Query:

- Import raw CSV/Excel file into Power BI.
- Remove blank and duplicate rows.
- Replace null amounts with 0 or remove invalid rows.
- Convert **date-time** to separate **Date** and **Hour** columns.

- Create new columns such as:
 - **Amount Range** (Low, Medium, High)
 - **Time Slot** (Morning, Afternoon, Evening, Night)

3.3 Data Modeling (Power BI & DAX)

- Create a **Fact Table** for transactions and **dimension tables** for:
 - Date
 - Customer
 - Merchant
 - Location
- Establish relationships between tables.
- Define **DAX measures**, such as:
 - Total Transactions
 - Total Fraud Transactions
 - Fraud Percentage
 - Total Fraud Amount
 - Average Transaction Amount

3.4 Dashboard Designing Process

Steps followed:

1. Identify **KPIs** to display on the dashboard.
2. Choose suitable **charts** for each analysis (bar, line, donut, map).
3. Arrange visual elements on the report page in a **clean layout**.
4. Add **slicers** for filters like Date, Country, Merchant Category, Fraud Flag.
5. Format visuals with proper **titles, labels, legends, and tooltips**

CHAPTER 4: SYSTEM DESIGN

4.1 Tools & Technologies Used

- **Power BI Desktop** – main dashboard tool.
- **Power Query** – data cleaning and transformation.
- **DAX** – calculations and measures.
- **Microsoft Excel / CSV** – source data.
- **PowerPoint / Word** – report and presentation.

4.2 Data Flow Diagram (Conceptual Description)

1. Input: Transaction dataset in CSV/Excel.
2. Power Query: Clean and transform data.
3. Data Model: Build relationships and measures in Power BI.
4. Visualization Layer: Create charts, KPIs, maps.
5. User Interaction: Apply filters and analyze fraud trends.

4.3 Architecture of the Dashboard

- **Data Layer:** Transaction dataset.
 - **Processing Layer:** Power Query + DAX.
 - **Visualization Layer:** Charts, tables, KPI cards, slicers.
 - **User Layer:** Analyst/manager using the dashboard to view fraud risk.
- .

CHAPTER 5: IMPLEMENTATION

5.1 KPI (Key Performance Indicator) Setup

Main KPIs used:

- **Total Transactions**
- **Total Fraud Transactions**
- **Fraud Percentage**
- **Total Fraud Amount (₹)**
- **Average Fraud Amount (₹)**

Each KPI is shown using **Card visuals** in Power BI.

5.2 Fraud Distribution by Transaction Type

- Visual: **Donut / Pie Chart**
- Categories: Online, POS, ATM, Contactless, International, etc.
- Purpose: To see which transaction type has the **highest fraud share**.

5.3 Fraud Analysis by Amount Range

- Visual: **Bar Chart**
- Amount Ranges:
 - 0–1000
 - 1001–5000
 - 5001–10000
 - Above 10000
- Helps to identify whether fraud is more common in **small, medium, or high-value** transactions.

5.4 Time-Based Fraud Analysis

- Visual: **Line or Column Chart**
- X-axis: Time of day or date.
- Y-axis: Number of frauds.
- Shows **peak hours** or **peak days** when fraud is more frequent.

5.5 Customer Segment & Location-Based Risk

- Visual: **Map / Bar Chart**
- Shows fraud count per **country, state, or city**.
- Also can show risk by **customer segment** (e.g., new customers, high spenders).

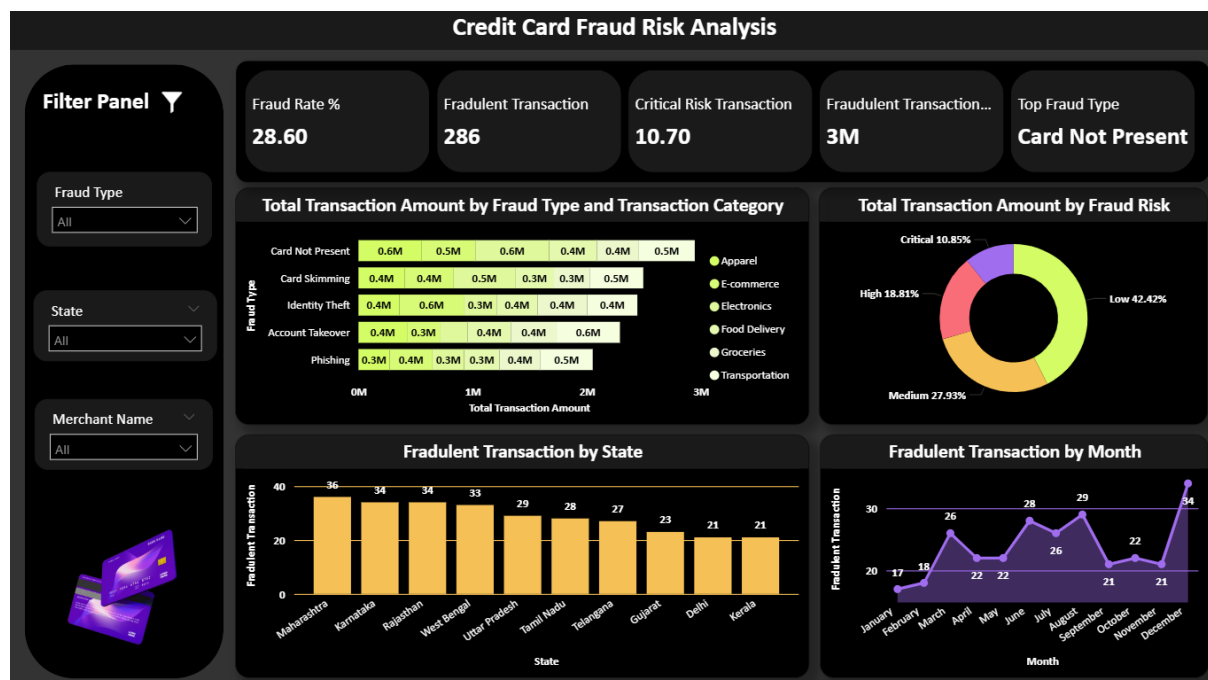
CHAPTER 6: RESULTS & ANALYSIS

6.1 Dashboard Output Screenshots (Explain in words)

This section presents the visual representation of the **Credit Card Fraud Risk Analysis** carried out using **Microsoft Power BI**. The dashboard contains multiple KPIs and analytical charts that help in understanding fraud trends, risk levels, transaction categories, and geographic distribution.

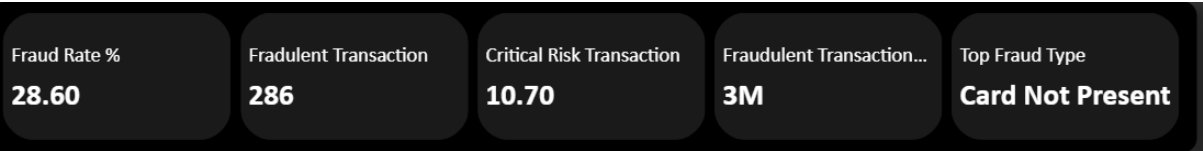
The following figures highlight different aspects of credit card fraud patterns, including fraud types, risk segmentation, monthly variation, and state-wise distribution

Figure 6.1: Complete Fraud Analysis Dashboard Overview



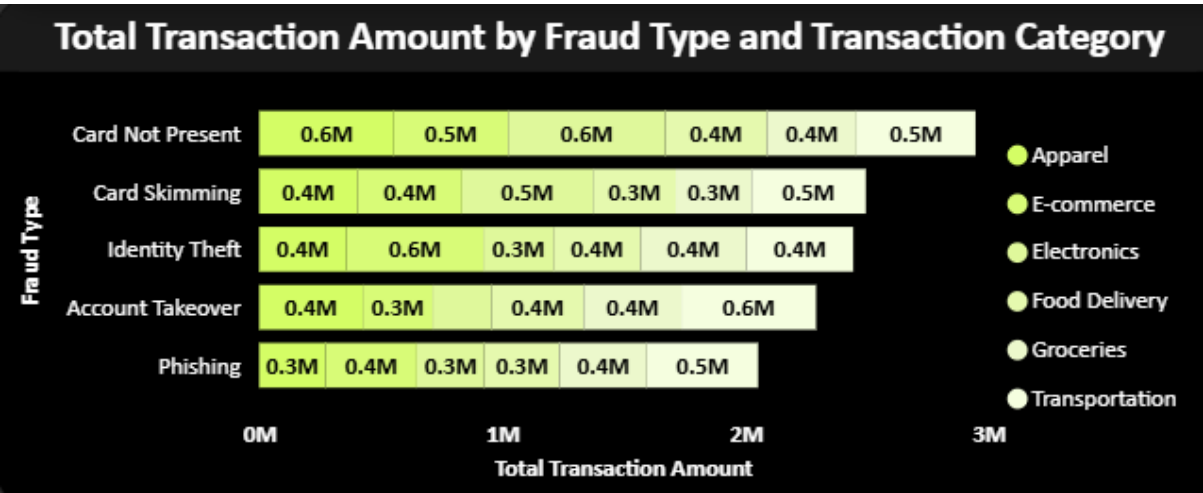
This figure displays the overall dashboard layout showing filters, KPIs, charts, and visual analytics. It gives a complete overview of fraud rate, total fraudulent transactions, and top fraud types detected in the dataset

Figure 6.2: Fraud Detection KPI Summary



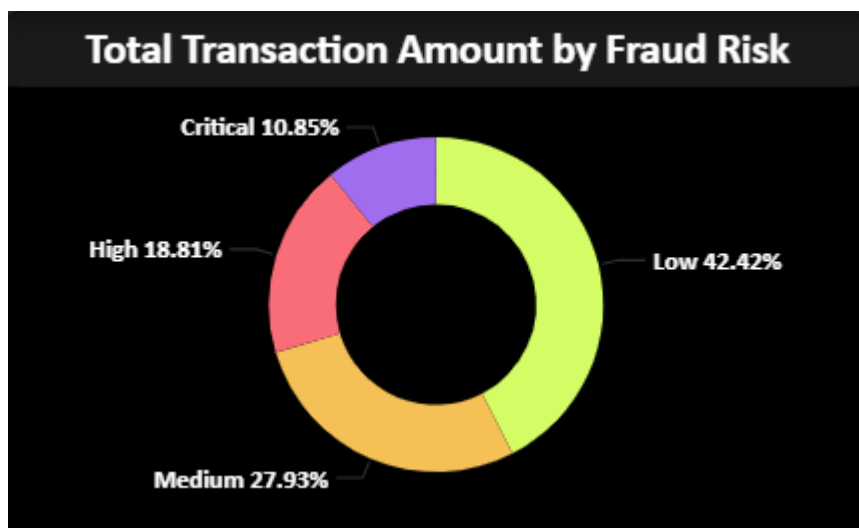
This chart showcases the major KPIs including Fraud Rate %, Total Fraudulent Transactions, Critical Risk Transactions, Total Fraudulent Amount, and Top Fraud Type. These KPIs provide a quick summary of the fraud status and overall risk level.

Figure 6.3: Fraud Type vs Transaction Category



This visual compares different fraud types—such as Card Not Present, Card Skimming, Identity Theft, Account Takeover, and Phishing—across various transaction categories like Apparel, Electronics, Groceries, E-commerce, and Transportation. It helps identify which fraud type is most frequent in specific purchasing categories.

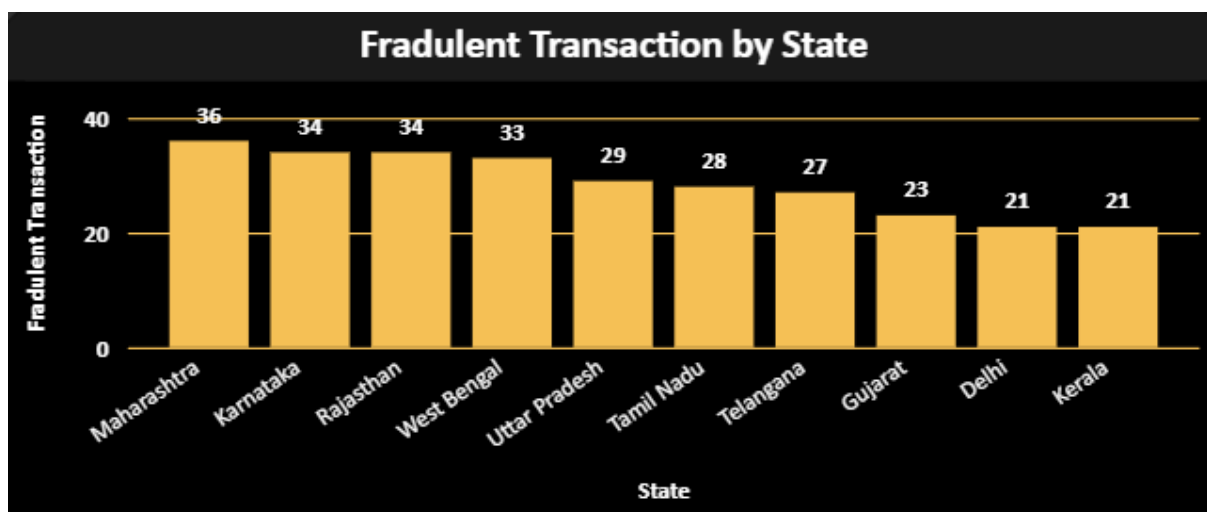
Figure 6.4: Fraud Risk Level Distribution



This donut chart divides the fraudulent transactions into four major risk levels: Low, Medium, High, and Critical.

It highlights that while low-risk transactions are most common, high and critical risk transactions form a significant portion and require immediate attention.

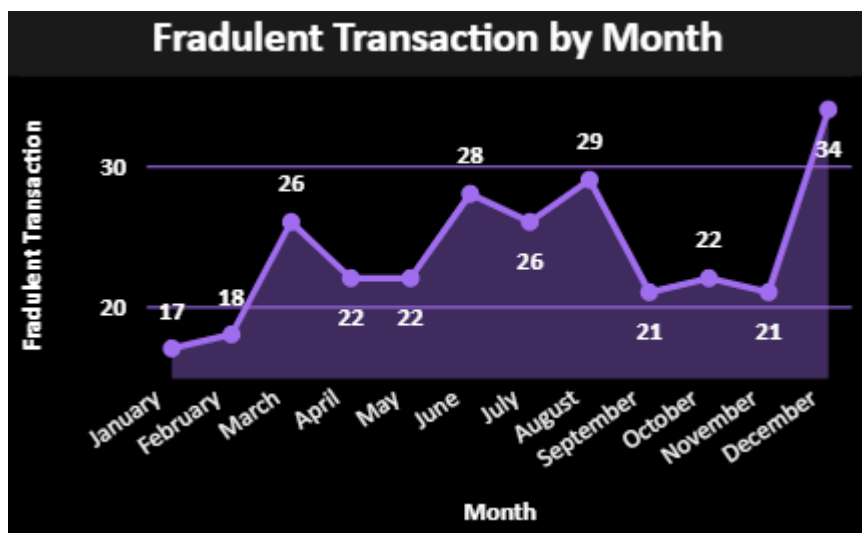
Figure 6.5: Fraudulent Transactions by State



This chart displays the number of fraudulent transactions recorded in different states such as Maharashtra, Karnataka, Rajasthan, West Bengal, and Uttar Pradesh.

It helps identify which states are more prone to card fraud activities.

Figure 6.6: Monthly Trend of Fraudulent Transactions



This line chart shows the monthly fluctuation of fraudulent transactions. The trend indicates peaks during mid-year months and a significant high in December, suggesting increased risk during festive and holiday seasons.

6.2 Insights / Observations

1. Fraud Rate is High:

The dashboard shows a fraud rate of 28.60%, indicating that a large portion of total transactions are fraudulent.

2. Card Not Present is the Top Fraud Type:

The most common fraud type is Card Not Present, which usually occurs in online or remote transactions.

3. Total Fraud Amount is Very Large:

The total fraudulent transaction amount is 3M, showing a major financial loss risk.

4. Fraud Cases are Higher in Certain States:

States like Maharashtra, Karnataka, Rajasthan, and West Bengal have the highest number of fraudulent transactions.

5. Monthly Fraud Trend Shows Peaks:

Fraud cases are lower in January but significantly increase in July, August, and December.

6. Risk Level Distribution is Clearly Defined:

Fraud is divided into Low, Medium, High, and Critical risk levels, with Low being the highest percentage, but High and Critical risks still being concerning.

CHAPTER 7: CONCLUSION & FUTURE SCOPE

7.1 Conclusion

The **Credit Card Fraud Risk Analysis** project successfully demonstrates how **Power BI** can be used to transform raw transaction data into **interactive visual insights**. By cleaning, modeling, and visualizing the data, it becomes easier to:

- Understand fraud behavior.
- Identify risky transaction types and regions.
- Support decision-making for banks and financial institutions.

This work proves that even without heavy coding, **BI tools** can play an important role in **fraud monitoring systems**.

7.2 Limitations

- Uses **historical data only**; no real-time streaming.
- The dataset may be **synthetic or limited in size**.
- The project focuses on **visual analytics**, not deep machine learning prediction.

7.3 Future Scope

- Integrate **real-time transaction data** using IIoT and API-based streaming.
- Apply **machine learning models** for automatic fraud prediction.
- Add **alert systems** that notify when risk exceeds a threshold.
- Deploy the dashboard as a **web or mobile app** for bank officers.

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ANNEXURES / APPENDIX

Appendix A – Raw Dataset (Excel/CSV File)

- The raw dataset used for this project contains **credit card transaction records**, which include both genuine and fraudulent transactions.
 - It consists of attributes such as **Transaction ID, Customer ID, Amount, Merchant Category, Transaction Type, Date & Time, State, and Fraud Flag (0 = Non-Fraud, 1 = Fraud)**.
 - The dataset was stored and processed in **Microsoft Excel / CSV format** before data cleaning and visualization.
 - Each record represents a single credit card transaction and forms the foundation for all fraud-risk analysis performed in Power BI.
 - The dataset was imported into **Power BI** for transformation, modeling, and dashboard creation.
-

Appendix B – Data Preparation / Fake Dataset Logic (If Dataset is Synthetic)

Since detailed real-world credit card fraud datasets are not publicly available due to privacy and banking security concerns, a **synthetic dataset** was prepared for analytical and academic purposes.

The fake dataset was designed to mimic real financial transaction behavior and commonly observed fraud patterns.

Data Generation Logic:

1. Base Transaction Count:

A starting transaction count was assumed and expanded with random variations to simulate real activity.

2. Fraud Percentage Logic:

- Fraud Rate set between **1% – 30%**, depending on category and risk type.

- Fraud Flag assigned using probability-based random distribution.

3. Amount Range Distribution:

- Low Value: 100 – 1000
- Medium Value: 1001 – 5000
- High Value: 5001 – 20000

Higher value transactions have a slightly higher probability of being classified as fraud.

4. Fraud Type Logic:

Fraud types were distributed based on realistic financial patterns:

- Card Not Present
- Card Skimming
- Identity Theft
- Account Takeover
- Phishing

5. State Distribution:

Fraud cases were distributed across Indian states such as **Maharashtra, Karnataka, Rajasthan, West Bengal, and Uttar Pradesh**, according to typical digital-fraud trends.

6. Timestamp Logic:

Transactions were assigned random dates and times to generate a realistic **monthly trend** and identify fraud spikes.

7. Rounding and Validation:

All values were validated and formatted properly before using the dataset for dashboard creation.

Appendix C – Final Dashboard Layout

This appendix presents the final layout of the **Credit Card Fraud Risk Analysis** dashboard created in **Microsoft Power BI**.

The dashboard includes:

- **KPI Cards:** Fraud Rate, Total Fraud Transactions, Critical Risk %, Total Fraud Amount, Top Fraud Type.
- **Charts:**
 - Fraud Type vs Transaction Category
 - Fraud Risk Distribution (Donut Chart)
 - State-wise Fraud Cases
 - Monthly Fraud Trend
- **Filters / Slicers:** Fraud Type, State, Merchant Name
- **Visual Theme:** Dark background theme for better clarity and contrast.

